EXPERIMENT 3

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PROGRAM:

```
clc;
clear;
close all;

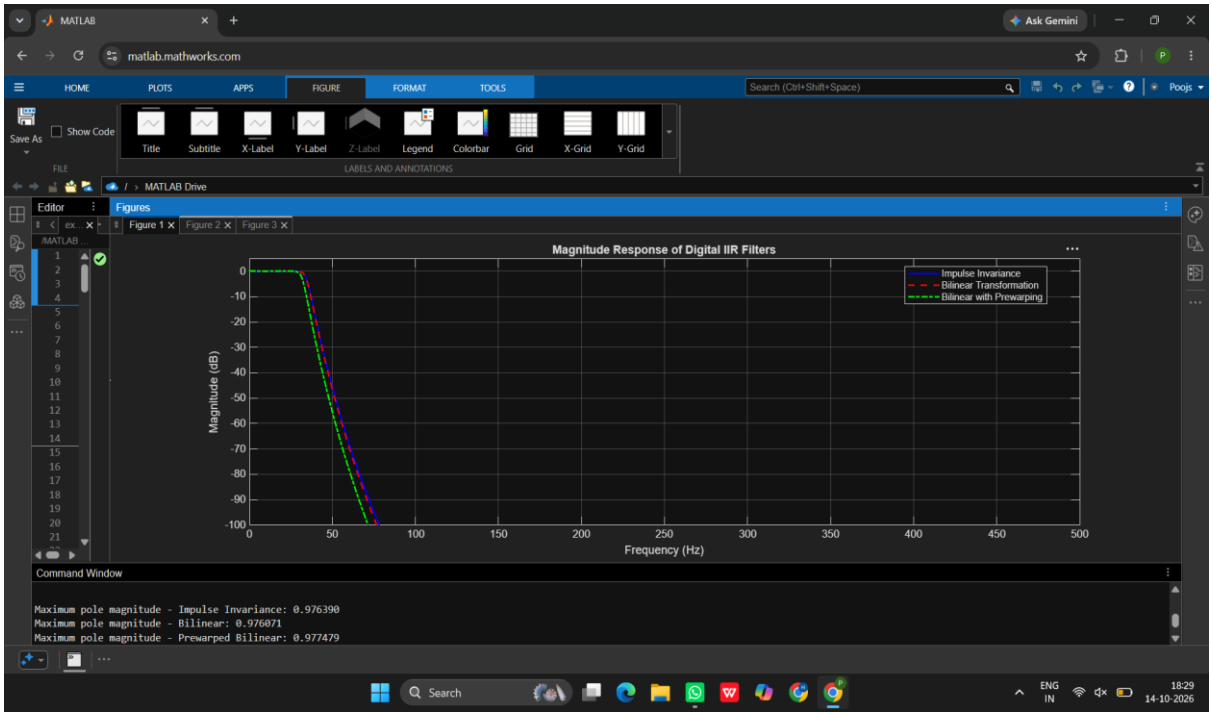
%% Input Specifications
Fs = 1000;        % Sampling frequency in Hz
T = 1/Fs;         % Sampling period

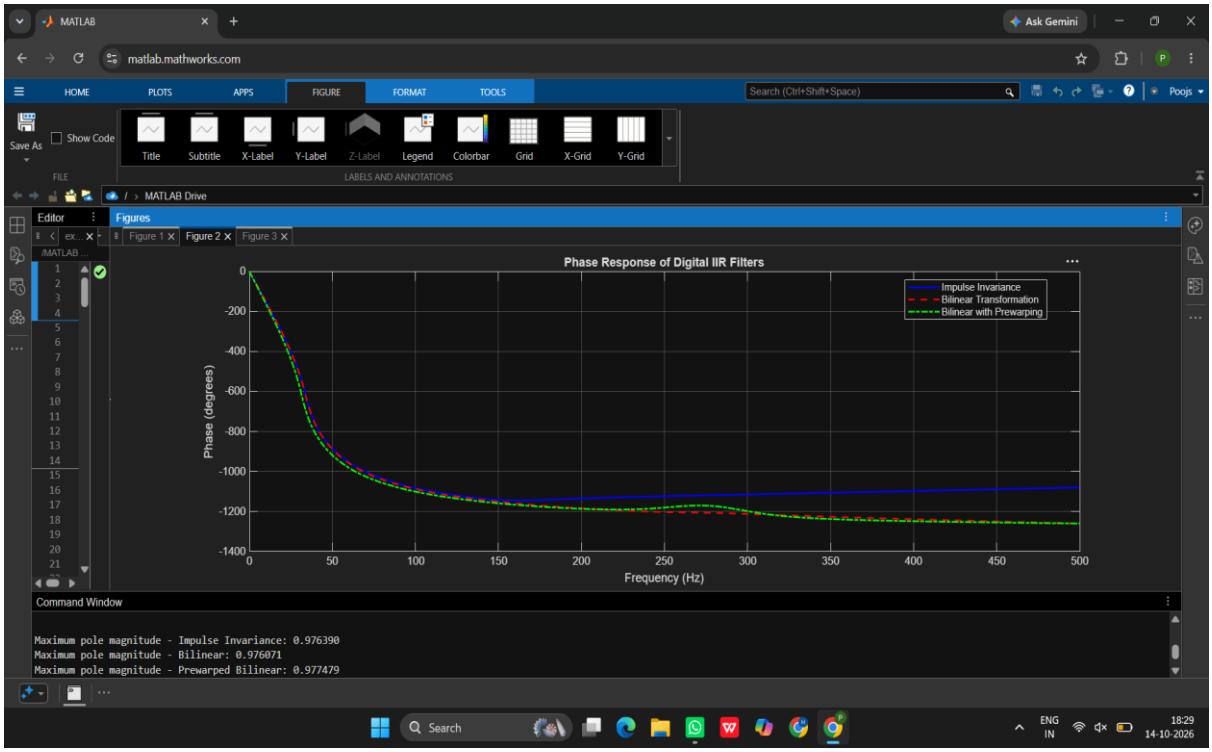
Fp = 200;         % Analog passband edge in rad/s
Fstop = 300;      % Analog stopband edge in rad/s
...fprintf('\nFilter Order: %d\n', N);

fprintf('Number of Impulse Invariance numerator coefficients: %d\n', ...
   length(bII));

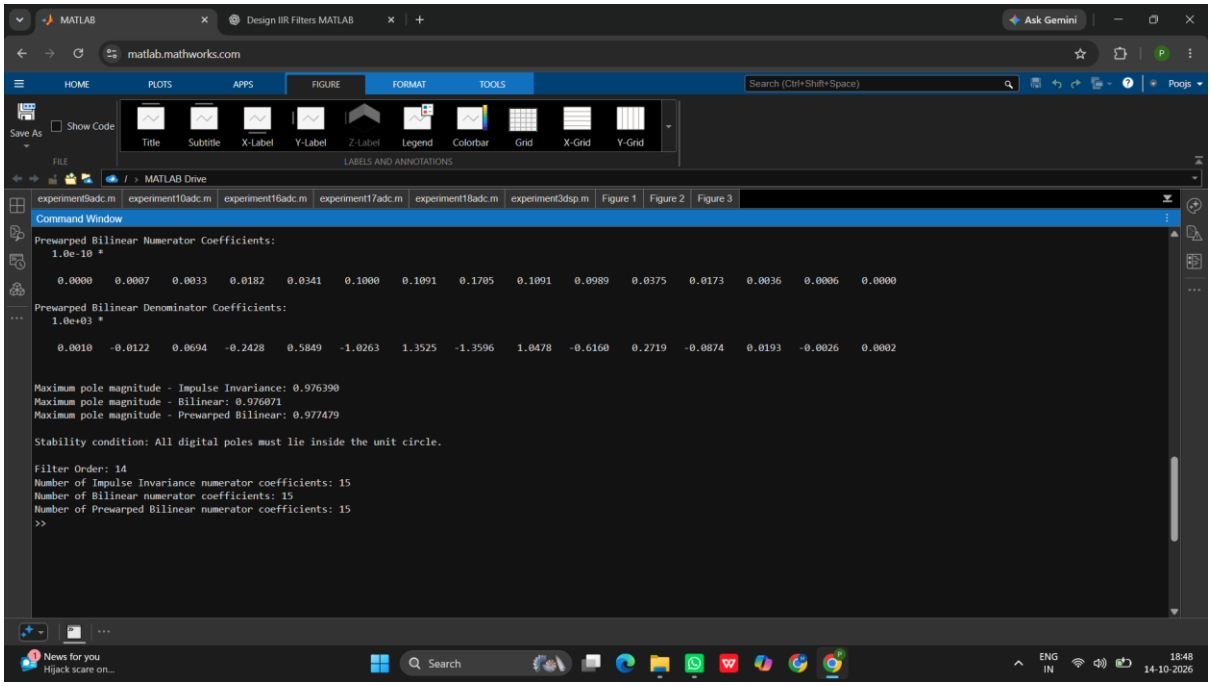
fprintf('Number of Bilinear numerator coefficients: %d\n', ...
   length(bBL));

fprintf('Number of Prewarped Bilinear numerator coefficients: %d\n', ...
   length(bPW));
```


Magnitude Response of Digital IIR Filters
Impulse Invariance
Bilinear Transformation
Bilinear with Prewarping
Magnitude (dB)
Frequency (Hz)
Command Window
Maximum pole magnitude - Impulse Invariance: 0.976390
Maximum pole magnitude - Bilinear: 0.976071
Maximum pole magnitude - Prewarped Bilinear: 0.977479


Phase Response of Digital IIR Filters
Impulse Invariance
Bilinear Transformation
Bilinear with Prewarping
Phase (degrees)
Frequency (Hz)
Command Window
Maximum pole magnitude - Impulse Invariance: 0.976390
Maximum pole magnitude - Bilinear: 0.976071
Maximum pole magnitude - Prewarped Bilinear: 0.977479


Frequency Warping in Bilinear Transformation
Digital Frequency (Hz)
Equivalent Analog Frequency (Hz)
Bilinear Frequency Mapping
Linear Frequency Mapping
Command Window
Maximum pole magnitude - Impulse Invariance: 0.976390
Maximum pole magnitude - Bilinear: 0.976071
Maximum pole magnitude - Prewarped Bilinear: 0.977479


Command Window
Prewarped Bilinear Numerator Coefficients:
1.0e-10 *
0.0000 0.0007 0.0033 0.0182 0.0341 0.1000 0.1091 0.1705 0.1091 0.0989 0.0375 0.0173 0.0036 0.0006 0.0000
Prewarped Bilinear Denominator Coefficients:
1.0e+03 *
0.0010 -0.0122 0.0694 -0.2428 0.5849 -1.0263 1.3525 -1.3596 1.0478 -0.6160 0.2719 -0.0874 0.0193 -0.0026 0.0002
Maximum pole magnitude - Impulse Invariance: 0.976390
Maximum pole magnitude - Bilinear: 0.976071
Maximum pole magnitude - Prewarped Bilinear: 0.977479
Stability condition: All digital poles must lie inside the unit circle.
Filter Order: 14
Number of Impulse Invariance numerator coefficients: 15
Number of Bilinear numerator coefficients: 15
Number of Prewarped Bilinear numerator coefficients: 15
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